

Experiment ‘p_1_8_painting’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_8_painting

Natural language statement: null

Method signature: p_1_8_painting() returns (s: string)

Ensures

- `s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

Functional Code Given

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_8_painting()` returns `(s: string)`

The method should respect the following contract:

`ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@\n", 5)`

The contract uses the following dafny code:

```
function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2HPySD71FNMvIBsfM6ZOgNXB8oV

Model: gpt-5.5-2026-04-23

Created at: 1780579207

Finish reason: stop

Usage: 378 tokens in, and 493 tokens out

```
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@\n" + ""))));
}
```

Final Program

```

function mystring(s: string, n: nat): string
{
  if n == 0 then "" else s + mystring(s, n - 1)
}

method p_1_8_painting() returns (s: string)
  ensures s == mystring(" !!!@@@@@@@@@@@@@@@@@@@@@\\n", 5)
{
  s := " !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" +
    (" !!!@@@@@@@@@@@@@@@@@@@@@\\n" + "")))));
}

```

Total Token Usage

Input tokens: 378

Output tokens: 493

Reasoning tokens: 398

Sum of ‘total tokens’: 871

Experiment Timings

Overall Experiment started at 1780579206875, ended at 1780579265999, lasting 59124ms (59.12 seconds)

Iteration #1 started at 1780579206883, ended at 1780579265999, lasting 59116ms (59.12 seconds)